Proceedings of the Design Society, Volume 6: DESIGN 2026
https://doi.org/10.1017/pds.2026.10677

Engineering design and entrepreneurship education: insights from the Design Society

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ABSTRACT: The Design Society unites researchers and practitioners to advance innovative engineering design. As entrepreneurship grows in importance, integrate design principles into its education seems necessary. This paper reviews Design Society publications to map how entrepreneurship, education, and sustainability intersect, it identifies the Design Society's contributions in the field to propose strategies and future research to strengthen its influence. It highlights the possible role of the Design Society in leading education advancements in entrepreneurship and sustainability.

KEYWORDS: design, entrepreneurship, sustainability, design education, engineering design

1. Introduction

Entrepreneurship is considered a significant driver of economic growth, innovation, and societal progress (Cohen & Winn, 2007; Naude, 2008). It fosters job creation, technological advancements, and can support escaping economic recessions, and its importance has been consistently demonstrated across various disciplines (Barba-Sánchez & Atienza-Sahuquillo, 2018). Even more so, it has the sustainable entrepreneurship can mitigate market imperfections leading to environmental degradation and support structural transition processes from low-income to high-income (Cohen & Winn, 2007; Naude, 2008). Central to this transformative power is entrepreneurship education, which equips individuals with the skills and knowledge to navigate complex challenges and translate ideas into impactful solutions (OECD, 2022; Pittaway & Cope, 2007). At the intersection of entrepreneurship and engineering design lies immense potential for groundbreaking innovations that address pressing global challenges, particularly in sustainability (Naude, 2008; Qian et al., 2024). Entrepreneurship and engineering have a special connection, one that gets growing attention in volatile economic times (Jarrar & Anis, 2017). Moreover, with the growing need for green skills in the job market to enable the green transition (Weiss et al., 2024), engineering entrepreneurship education often follows an engineering curriculum and focuses on developing skills related to customer discovery, ideation, and innovation driven by technical precision and rapid prototyping (Jarrar & Anis, 2017; Fujimaki et al., 2025). Traditional approaches that prioritize speed-to-market often fail to incorporate sustainability during the early stages of product development (Jamil et al., 2024). Given the escalating environmental challenges, greater emphasis must be placed on fostering sustainable entrepreneurship (Cohen & Winn, 2007). Engineering-focused entrepreneurship education must integrate sustainability as a fundamental principle to ensure that emerging companies and their products are innovative, efficient, and environmentally responsible. For instance, programs like Technology Entrepreneurship for Engineers and Computer Scientists at the University of Ottawa effectively teach vital entrepreneurial skills but lack a comprehensive sustainability perspective (Jarrar & Anis, 2017). Engineers engaged in product development possess the technical expertise necessary to design products that are "sustainable-by-default," leveraging renewable materials, circularity, and

https://doi.org/10.1017/pds.2026.10677 Published online by Cambridge University Press

minimal environmental impact. By applying Design-for-X principles, they can enhance manufacturability and optimize the overall environmental footprint of their products (Reddy et al., 2023). Encouraging the alignment of entrepreneurial activities with sustainable development goals is critical (Li & Seering, 2019; Reddy et al., 2023). The Design Society, an interdisciplinary community dedicated to advancing design methods and sustainable practices, is uniquely positioned to drive this change. Its mission to influence sustainable innovation provides a strong foundation to advance practices on sustainable engineering design into entrepreneurial education. However, a systematic review of the Design Society's online publication database uncovered notable gaps in its engagement in sustainability in connection with entrepreneurship and entrepreneurship skills. Among 76 papers focusing on entrepreneurship, only three explicitly addressed sustainability. Additionally, 38 papers explored the intersection of entrepreneurship and education, and 21 examined entrepreneurship within the context of product development. Yet none explicitly dealt with circularity or the circular economy in relation to entrepreneurship. Circularity matters in this context because it shifts value creation to designing products that keep materials in circulation (e.g., that are actively reuse, repair, remanufacturing), which can substantially reduce lifecycle impacts and increase sustainability. This reveals a significant oversight in aligning engineering design with sustainability principles and practices within entrepreneurial education. This gap represents a missed opportunity for the Design Society, which is uniquely positioned to lead advancements in sustainable engineering design. Despite its interdisciplinary expertise and influential role in design research, the absence of a dedicated focus on entrepreneurial education - both in its conferences and thematic programming - limits its potential to foster sustainable innovation. This omission underscores the need for a more integrated approach that bridges the gap between engineering design, sustainability, and entrepreneurship. To address this gap, our study examines the Design Society's contributions to entrepreneurship and recommends strategies for embedding sustainability as a core principle within entrepreneurial education.

2. Methodology and data analysis

2.1. Methodology

The methodological approach of this research centered around a comprehensive review of all available publications in the Design Society's digital library, which includes conferences such as International Conference of Engineering Design (ICED), Norddesign, and Design among others. By aggregating the available data from these diverse conferences and publications, we aimed to capture an overall perspective of the Design Society's discourse on entrepreneurship and entrepreneurial education. The decision to include the full range of publications ensured access to peer-reviewed, high-quality research reflecting the broader perspectives within the design community. Our systematic search used "entrepreneurship" as the primary search term across the whole database, including abstracts, titles, keywords, and content. This initial search yielded 76 articles. After data validation, four articles were found to be corrupted or not available and consequently excluded, leaving 71 articles for initial screening. Upon detailed review, 57 articles met our inclusion criteria and were selected for comprehensive analysis. These articles were systematically analyzed using a structured spreadsheet that captured key dimensions, including research context, main findings, challenges, opportunities, and common themes. This documentation enabled us to identify recurring patterns and emerging themes in design entrepreneurship research, particularly in educational contexts, which informed our subsequent in-depth analysis.

2.2. Data analysis

Our analysis focused on education and design skills development as the key lens in entrepreneurship, a choice well-supported by the thematic emphasis in the literature. A thorough review indicates that around 36 papers (over 50% of the sample) specifically examine educational, skills development, or capacity building elements in entrepreneurship. The keyword analysis reveals that "entrepreneurship" is mentioned in 25 papers, while educational and skills-related terms feature in 16 papers as central themes. Notably, research that is focusing on curriculum development (9 papers), skills development (10 papers), and capacity building (6 papers) highlights the dominant role that educational frameworks and pedagogical methods play in the field. The geographic distribution illustrates significant contributions from Nordic and Western European institutions, with Danish institutions involved in 8 papers, UK

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institutions in 6 papers, and Swedish institutions in 5 papers, indicating a strong Northern European influence on design entrepreneurship education research within the Design Society. The recurring use of terms such as "innovation" (13 papers) and "design thinking" (11 papers) within educational settings underscores that academic institutions act as vital environments for integrating design and entrepreneurial skills. This strong emphasis on education, skills development, and capacity building in the literature supports our educational-focused analytical method.

3. Theoretical frame of reference

3.1. Benefits and outcome

The integration of entrepreneurship into design education has found yielding numerous benefits for students, educators, and institutions. A recurring theme across the reviewed literature is the enhancement of creativity, innovation, and problem-solving skills among students exposed to entrepreneurial learning experiences (Berglund et al., 2011; Effa et al., 2011; Esparza et al., 2019; Harsaae et al., 2023; Jerkku et al., 2016; Kirjavainen et al., 2016; Lynch et al., 2016). These studies highlight how the introduction of business-oriented thinking and real-world challenges pushes designers to think beyond purely aesthetic or functional considerations, encouraging them to develop more holistic and market-driven solutions. Several papers also report improved market-readiness and real-world applicability of student projects when entrepreneurship is integrated into the design curriculum (Elaver, 2015; Fujimaki et al., 2025; Gheorghe et al., 2015; Petersen, 2017; Shin, 2008). By engaging with industry partners, working on live briefs, and considering the commercial viability of their ideas, students gain valuable insights into the realities of the business world and are better prepared for the challenges of professional practice. This exposure can lead to successful commercialization of student projects and even the creation of start-ups, as evidenced in some case studies (De Vere, 2014; Elaver, 2015; Gheorghe et al., 2015). Moreover, the reviewed literature suggests that entrepreneurship education can foster an entrepreneurial mindset and increase entrepreneurial intentions among design students (Germany & Lillegard, 2017; Håkansson & Holmquist, 2011; Sigurjónsson & Boks, 2011; Yu et al., 2019). By providing students with the tools, knowledge, and confidence to pursue their own ventures, these programs can contribute to the development of a more innovative and enterprising design workforce (Germany & Lillegard, 2017; Sigurjónsson & Boks, 2011). Some studies even indicate that exposure to entrepreneurship education can lead to higher rates of self-employment and business creation among design graduates (Håkansson & Holmquist, 2011; Yu et al., 2019).

3.2. Educational approaches and best practices

The reviewed literature showcases a diverse range of educational approaches and pedagogical strategies for integrating entrepreneurship into design education and vice versa. One prominent theme is the emphasis on interdisciplinary collaboration and cross-disciplinary integration (Baelus et al., 2013; Germany & Lillegard, 2017; Pavel & Berg, 2014). Many programs seek to bridge the traditional divide between design and business education by creating opportunities for students from different backgrounds to work together on projects and share their expertise (Baelus et al., 2013; Germany & Lillegard, 2017). This can take the form of joint courses, collaborative workshops, or even the creation of new hybrid programs at the intersection of design and entrepreneurship (Baelus et al., 2013; Pavel & Berg, 2014). Another key trend is the adoption of experiential, project-based learning methodologies that engage students in real-world challenges and hands-on problem-solving (Berglund et al., 2011; Effa et al., 2011; Elaver, 2015). Rather than relying on theoretical lectures or case studies, these approaches immerse students in the entrepreneurial process, from ideation and market research to prototyping and pitching (Berglund et al., 2011; Shin, 2008). By working on live projects with industry partners or developing their own ventures, students gain practical skills and experience that can be directly applied in their future careers (Elaver, 2015; Gheorghe et al., 2015). Design thinking emerges as a central framework in many of the reviewed papers, with educators leveraging its human-centered, iterative approach to help students navigate the complexities of entrepreneurial problem-solving (Harsaae et al., 2023; Lynch et al., 2017; Nielsen et al., 2013; Sánchez Ruano et al., 2025). Other popular methodologies include the lean startup approach (Lynch et al., 2016), effectuation theory, and business model innovation (Nielsen et al., 2013). Framework like this provide students with structured yet flexible tools for testing assumptions, gathering user feedback, and rapidly prototyping solutions. Successful programs also tend to incorporate a strong

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element of industry collaboration and professional networking (Elaver, 2015; Gheorghe et al., 2015; Shin, 2008). By bringing in guest speakers, mentors, and project partners from the business world, educators can expose students to different perspectives, provide role models, and help them build valuable connections for their future careers (Elaver, 2015; Shin, 2008). Some programs even go a step further by integrating internships or co-op placements into the curriculum, allowing students to gain real-world experience and apply their learning in professional settings (Petersen, 2017). However, implementing these approaches also requires careful consideration of factors such as the institutional culture, available resources, and faculty expertise. Several papers highlight the challenges of integrating entrepreneurship into traditional design programs, including resistance to change from both students and educators (Hamilton & Brisco, 2023; Watkins & Clarke, 2018), lack of appropriate teaching materials and case studies (Yu et al., 2019), and difficulties in assessing learning outcomes (Effa et al., 2011).

3.3. Challenges

Despite the numerous benefits and successful approaches identified in the literature, integrating entrepreneurship into design education is not without its challenges. One recurring issue is the difficulty of fostering genuine cross-disciplinary collaboration and communication between students and faculty from different backgrounds (Baelus et al., 2013; Buchert & Stark, 2018; Germany & Lillegard, 2017). While the ideal of interdisciplinary teamwork is widely praised, the reality is often tainted by conflicts arising from differing disciplinary cultures, language, and expectations (Pavel & Berg, 2014). The diversity challenge can be addressed by increasing participant diversity through actively grouping students from various academic backgrounds (Sugiue et al., 2024). Another significant challenge is the resistance to change from both students and educators who may be entrenched in traditional design education models (Hamilton & Brisco, 2023; Watkins et al., 2021). For students, the addition of entrepreneurship content can also be seen as a distraction from their core design studies or an unwelcome burden on their already heavy workloads (Hamilton & Brisco, 2023). Educators, on the other hand, may lack the necessary expertise, confidence, or motivation to integrate entrepreneurship into their teaching, particularly if they do not have prior business experience or training (Watkins & Clarke, 2018). This resistance can be compounded by institutional barriers, such as rigid curricular structures, limited resources for faculty development, and a lack of recognition for interdisciplinary teaching efforts (Hamilton & Brisco, 2023; Yu et al., 2019).

Several papers also highlight the ongoing debate around the appropriate balance between design and entrepreneurship skills in the curriculum (Crea, 2014; De Vere, 2013; Maguire, 2006; Mejia & Betancourt, 2013). While some argue for the need to equip design students with a broad set of business competencies to succeed in today's competitive job market (Maguire, 2006; Mejia & Betancourt, 2013), others caution against diluting the core strengths and values of design education in pursuit of market relevance (De Vere, 2013; Sánchez Ruano et al., 2025). A related issue is the potential misalignment between the goals and priorities of academia and industry when it comes to preparing design entrepreneurs (Elaver, 2015; Petersen, 2017). While universities may focus on fostering critical thinking, creativity, and social responsibility, businesses often prioritize more tangible outcomes, such as technical skills, financial acumen, and market readiness (Petersen, 2017). Bridging this gap requires ongoing dialogue and partnership between educators and industry stakeholders to ensure that curricula remain relevant and responsive to evolving professional demands (Elaver, 2015). Finally, the reviewed literature reveals a lack of consensus around effectively assessing and measuring entrepreneurship education's impact on design students' learning outcomes and career trajectories (Effa et al., 2011; Mejia & Betancourt, 2013). This challenge is further highlighted in research examining how educational activities before, during, and after internships contribute to students' entrepreneurial competence (Petersen, 2017). Traditional design assessment methods, such as portfolios and critiques, may not fully capture the business skills and entrepreneurial mindsets developed through these programs (Effa et al., 2011; Mejia & Betancourt, 2013). Recent studies on self-efficacy and motivation in entrepreneurship education suggest that assessment should consider both psychological and skill-based outcomes (Jerkku et al., 2016; Lynch et al., 2016). The importance of measuring cross-disciplinary skills development has also emerged as a crucial factor when assessing entrepreneurship education programs (Hamilton & Brisco, 2023). Moreover, tracking graduates' long-term entrepreneurial activities and success remains a methodological challenge, given the complexity of factors influencing career paths and the difficulty of attributing outcomes to specific educational interventions (Mejia & Betancourt, 2013). Traditional measures like employment rates may not effectively capture entrepreneurial outcomes, suggesting more

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nuanced metrics are needed (Watkins & Clarke, 2018). Developing robust, multi-dimensional assessment frameworks that balance creative, technical, and business competencies is an ongoing area of research and experimentation in the field (Effa et al., 2011; Mejia & Betancourt, 2013).

4. Research agenda and gaps

While the reviewed literature provides valuable insights into the integration of entrepreneurship and design education, it also highlights several areas where further research is needed to advance theory and practice within this domain. One key area is the need for more empirical studies that rigorously evaluate the impact of different pedagogical approaches and interventions on students' learning outcomes, entrepreneurial intentions, and career trajectories (Effa et al., 2011; Petersen, 2017; Yu et al., 2019). While many papers are describing innovative programs and dedicated curricula, only a few provide robust evidence of their effectiveness. Researchers started to address this gap. New analytical frameworks, like the LAD framework, are being developed to evaluate the learning process and depth. Additionally, long-term comparative studies are intended to measure the impact of interventions. The studies use pre/post assessments and post-program behavioral tracking to evaluate elements like diversity initiatives and design sprints (Sugiue et al., 2024; Fujimaki et al., 2025). More research is needed to identify best and effective practices, success factors, and contextual variables that influence entrepreneurship education outcomes in design settings. Another important avenue for future research is the exploration of how entrepreneurship education can be effectively adapted to different cultural, institutional, and disciplinary contexts (Baelus et al., 2013; Ekman et al., 2011; Keitsch, 2015).

While much of the literature has historically been based on case studies from North America and Europe, recent research is significantly broadening this perspective. New models, frameworks, and evaluations are emerging from diverse institutional and cultural contexts, including Japan, Mexico, and multicultural, multi-institutional programs based in Canada. (Keitsch, 2015 ; Fujimaki et al., 2025; Sugiue et al., 2024). Thus, this emerging body of work is already yielding valuable insights into the contextual factors, such as the need for structured cultural and educational management when managing multicultural student teams and the pedagogical challenges of bridging trans-disciplinary student backgrounds, including science, engineering, and liberal arts.

The rapid advancement of technology and the changing nature of work also present new opportunities and challenges for integrating entrepreneurship into design education (De Vere, 2013; Smulders et al., 2019). The growing importance of digital skills, remote collaboration, and lifelong learning in the design profession calls for research into how entrepreneurship curricula can harness emerging technologies and pedagogical innovations to better prepare students for the future of work (Hölttä-Otto et al., 2013; Smulders et al., 2019). Studies that explore the use of online platforms, virtual incubators, and blended learning models for fostering entrepreneurial competencies could offer valuable guidance for educators seeking to adapt to these evolving realities (De Vere, 2013; Tollestrup, 2011). Finally, there is a need for more research that takes a systemic and interdisciplinary perspective on the role of entrepreneurship education in driving sustainable development and social innovation (Baxter & Bruce, 2008; Jungersen & Hansen, 2014; Nielsen et al., 2013; Schaber et al., 2021). While some papers touch on the potential for designers to create positive social and environmental impact through entrepreneurship (Baxter & Bruce, 2008; De Vere, 2013; Krauss et al., 2024), there is scope for deeper exploration of how entrepreneurship education can be aligned with the United Nations' Sustainable Development Goals and integrated with emerging frameworks such as design for sustainability, circular economy, and social innovation (De Vere, 2013; Ekman & Ekman, 2009; Harsaae et al., 2023). Studies suggest that systems thinking and ecological perspectives are becoming increasingly crucial in design entrepreneurship education (Baxter & Bruce, 2008; Ekman & Ekman, 2009; Nielsen et al.,2013), with particular emphasis on developing solutions for rural and underserved communities (Ekman & Ekman, 2009; Keitsch, 2015). For example, the 'D+ Framework' is proposed as a model to move students from product-oriented solutions to systemic approaches to value creation by explicitly pairing strategic design with humanistic culture, foresight, and innovation (Sánchez Ruano et al., 2025). Integrating these approaches requires new pedagogical frameworks that can bridge traditional disciplinary boundaries (Baelus et al., 2013; Germany & Lillegard, 2017; Yu et al., 2019). Collaborative research integrating design, business, engineering, and social sciences can enhance understanding of how entrepreneurship education addresses complex global challenges (Pavel & Berg, 2014; Schaber et al., 2021; Simeone, 2014; Yu et al., 2019). While this paper emphasizes the role of education in embedding sustainability within entrepreneurship and entrepreneurship within design education, it is important to recognize that

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entrepreneurial and design decisions are shaped not only by competencies but also by economic and regulatory environments. As noted in prior studies, tensions often emerge between market readiness and broader societal responsibility (Petersen, 2017; De Vere, 2013). Even when entrepreneurship education fosters systems thinking and sustainability awareness (Ekman & Ekman, 2009; Harsaae et al., 2023), cost pressures and regulatory minimums may constrain implementation in practice. This suggests that multidisciplinary collaboration must extend beyond design and business education to include engagement with policymakers and regulatory actors. Entrepreneurship education should therefore be aligned with broader institutional frameworks and incentive structures, recognizing that meaningful transitions require coherence between academic reform, industry practice, and policy mechanisms.

5. Recommendations

Based on the insights from the reviewed literature and the collective expertise of the design community, we propose the following recommendations for strengthening design education that supports entrepreneurship.

Foster a culture of collaboration and interdisciplinarity: Design of programs and curricula should actively promote cross-disciplinary collaboration and partnerships, both within the institution and with external stakeholders such as business schools, engineering departments, and industry partners (Baelus et al., 2013; Hamilton & Brisco, 2023; Sivaloganathan & Shahin, 2013; Yu et al., 2019). This approach has been successfully demonstrated through joint courses, collaborative projects, faculty exchanges, and networking events that bring together diverse perspectives and expertise (L. S. Buck et al., 2017; Germany & Lillegard, 2017; Koh et al., 2016; Simeone, 2014; Watkins & Clarke, 2018). Research shows that such cross-disciplinary integration can enhance both technical and entrepreneurial competencies (Baelus et al., 2013; Effa et al., 2011; Lynch et al., 2017; Mejia & Betancourt, 2013). Programs that bridge academic disciplines have demonstrated particular success in developing innovative solutions while building business acumen (Hamilton & Brisco, 2023; Kirjavainen et al., 2016; Petersen, 2017; Shin, 2008). By cultivating a culture of openness, curiosity, and mutual respect, design education can break down silos and create fertile ground for entrepreneurial thinking and innovation (Andersen et al., 2013; Baelus et al., 2013; Pavel & Berg, 2014; Sivaloganathan & Shahin, 2013; Yu et al., 2019).

Integrate entrepreneurship throughout the curriculum: Rather than treating entrepreneurship as a separate course or module, design programs should seek to embed entrepreneurial learning experiences throughout the curriculum (De Vere, 2013; Fuller, 2017; Hölttä-Otto et al., 2013; Yu et al., 2019). This can involve infusing business concepts and skills into core design courses, creating opportunities for students to work on real-world projects with entrepreneurial elements (Lynch et al., 2017; Petersen, 2017), and providing elective courses or workshops that delve deeper into specific aspects of entrepreneurship such as market research, financial planning, and intellectual property (Lynch et al., 2017; Schaber et al., 2021; Shin, 2008; Smulders et al., 2019). Research demonstrates that integrated approaches lead to better learning outcomes than standalone entrepreneurship courses (Germany & Lillegard, 2017; Yu et al., 2019), particularly when combined with practical, hands-on experiences (Petersen, 2017; Tollestrup, 2011; Watkins & Clarke, 2018). The goal should be to create a seamless integration of design and entrepreneurship education that enhances, rather than detracts from, the development of creative problem-solving skills (Baelus et al., 2013; Hamilton & Brisco, 2023).

Embrace experiential and project-based learning: Design education is well-suited to entrepreneurial learning because of its emphasis on hands-on, experiential, and project-based pedagogies (Behrenbeck et al., 2020; Berglund et al., 2011; Lynch et al., 2016; Watkins & Clarke, 2018). Programs should double down on these approaches by creating more opportunities for students to engage in real-world challenges, work with industry partners, and develop their own ventures or products (Petersen, 2017; Quang Tran & Nhu Laursen, 2021). Research shows that practical experience through design clinics, incubators, or accelerators provides students with crucial resources, mentorship, and support to pursue entrepreneurial projects (Baelus et al., 2013; Jungersen & Hansen, 2014; Shin, 2008; Watkins & Clarke, 2018). Studies demonstrate the effectiveness of partnerships with local businesses, non-profits, or government agencies in providing students with authentic learning experiences (Jungersen & Hansen, 2014; Petersen, 2017; Watkins & Clarke, 2018). These real-world engagements have been shown to significantly enhance both technical and entrepreneurial capabilities (Hamilton & Brisco, 2023; Schaber et al., 2021), while building students' confidence in pursuing entrepreneurial opportunities (Jerkku et al., 2016; Lynch et al., 2017; Lynch et al., 2016). Evidence suggests that such experiential approaches lead to more successful transitions into professional practice and entrepreneurial ventures (Petersen, 2017; von Platen & Kitani, 2023).

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Leverage technology and digital platforms: The rapid advancement of technology is creating new opportunities for design education to support entrepreneurial learning (Böhmer & Lindemann, 2015; Li & Seering, 2019). Programs should explore how digital tools and platforms can be harnessed to facilitate remote collaboration, access to resources, and global connectivity (Fuller, 2017; Li & Seering, 2019). Studies demonstrate the growing importance of digital manufacturing technologies and open-source approaches in entrepreneurial education (Keitsch, 2015; Platen & Kitani, 2023; Yu et al., 2019). Research shows that digital platforms can effectively support prototyping, testing, and iteration of ideas (Esparza et al., 2019; Harsaae, Nielsen, et al., 2023), while also enabling new forms of collaborative innovation and knowledge sharing (Baxter & Bruce, 2008; Tollestrup, 2019; Yu et al., 2019). This can involve using online learning management systems to deliver entrepreneurship content, creating virtual incubators or maker spaces where students can prototype and test ideas, and leveraging social media and crowdfunding platforms to support student ventures (Böhmer & Lindemann, 2015; Li & Seering, 2019; Von Platen & Kitani, 2023). Evidence suggests that integrating these digital tools helps prepare students for emerging business models and entrepreneurial opportunities (Keitsch, 2015; Watkins & Clarke, 2018). By embracing technology, design education can expand its reach and impact while also preparing students for the digital realities of the entrepreneurial landscape (Watkins & Clarke, 2018; Yu et al., 2019).

Foster a global and sustainable mindset: Design education should strive to cultivate a global and sustainable mindset among students, recognizing the potential for entrepreneurship to drive positive social and environmental impact (Harsaae et al., 2023; Keitsch, 2015; Sørensen, 2020). This can involve integrating principles of sustainability, social innovation, and ethical leadership into the curriculum (Harsaae et al., 2023; Pavel & Berg, 2014a), while exposing students to diverse cultural perspectives and practices. Research demonstrates the importance of systems thinking and ecological approaches in developing sustainable solutions (Ekman & Ekman, 2009; Grimheden, 2009; Sørensen, 2020). Studies show that addressing real-world challenges in different cultural contexts enhances both innovation capabilities and social awareness (Bix, 2017; Germany & Lillegard, 2017; Sørensen, 2020). Programs that emphasize sustainable development have demonstrated success in preparing students to tackle complex global challenges (Harsaae et al., 2023; Keitsch, 2015), while building awareness of social responsibility in business creation. By fostering a sense of global citizenship and responsibility, design education can empower students to use their entrepreneurial skills for the greater good (Crea, 2014; Harsaae et al., 2023; Møller Nielsen et al., 2013).

Engage the design community as a catalyst for change: The design community has a critical role to play in advocating for and supporting the integration of entrepreneurship into design education (Baelus et al., 2013; Berglund et al., 2011). Professional associations, conferences, and publications should prioritize entrepreneurship as a key theme, showcasing best practices, success stories, and thought leadership in this area (De Vere, 2013; Germany & Lillegard, 2017). Research demonstrates that community engagement and professional networks significantly impact educational innovation (Baelus et al., 2013; Quang Tran & Nhu Laursen, 2021; Yu et al., 2019). Design educators and practitioners should actively participate in these forums, sharing their experiences, insights, and visions for the future of the field (Berglund et al., 2011; Lennart Sebastian Buck et al., 2017; Germany & Lillegard 2017). Studies show that strong industry-academia partnerships enhance both educational outcomes and professional development (Baelus et al., 2013; Koh et al., 2016). Evidence suggests that collaborative networks between institutions and practitioners drive innovative teaching practices and help establish new standards for design entrepreneurship education (Baelus et al., 2013; Maguire, 2006; Quang & Nhu, 2021). By collectively mobilizing around this agenda, the design community can create a powerful movement for change, influencing policies, standards, and practices in design education and beyond (Germany & Lillegard 2017; Pavel & Berg, 2014; Yu et al., 2019).

However, embedding sustainability within entrepreneurship education alone is insufficient. Meaningful transformation requires stronger integration of sustainability principles within legal frameworks, regulatory standards, and business ethics. In addition, accelerators, incubators, and venture capital actors should incorporate sustainability-related KPIs into funding and performance evaluation criteria. Aligning education, governance, and financing mechanisms is essential to create coherent incentives for genuinely sustainable entrepreneurial development.

6. Final remarks

The literature review highlights significant gaps in the integration of entrepreneurship into the Design Society's agenda, particularly in addressing sustainability within engineering-oriented entrepreneurial

https://doi.org/10.1017/pds.2026.10677 Published online by Cambridge University Press

education. While entrepreneurship is a critical driver of innovation and economic growth, the rapid development of products often neglects sustainability as a foundational principle. This oversight is especially critical in engineering contexts, where decisions made during design phases have far-reaching environmental and societal impacts. Analyzing conferences in one research community uncovers vital insights often overlooked by standard literature reviews, which mainly focus on journal articles (Snyder, 2019). Our focused examination of articles related to the Design Society proceedings highlight emerging trends, implementation hurdles, and practical solutions frequently absent in journal-centric reviews (Lisée et al., 2008). This method presents an alternative view to traditional systematic reviews by highlighting the rapid changes in engineering practices and applications within specialized networks.

The Design Society can lead a paradigm shift by embedding a "sustainable by default" paradigm in its programs. This involves equipping future engineers and entrepreneurs with the knowledge and tools to prioritize sustainability in the product lifecycle, utilizing renewable materials, fostering circularity, and minimizing environmental impacts. Interdisciplinary collaboration, experiential learning, and strong industry partnerships are essential for this vision. However, challenges like resistance to change, disciplinary silos, and curriculum balance need addressing. The Design Society can enhance its influence by integrating entrepreneurship educational research, creating dedicated tracks, workshops, and publications focusing on sustainability in engineering design with an entrepreneurship perspective. Future efforts should emphasize rigorous empirical research, adapt teaching methods, and leverage emerging technologies for sustainable development innovation. By taking these steps, the Design Society can play a pivotal role in shaping a future where design, engineering, and entrepreneurship work in tandem to create sustainable, impactful innovations. This will ensure that engineering design remains at the forefront of addressing global challenges while fostering a new generation of responsible, innovative leaders.

Finally, During the literature review, some articles in the Design Society database were corrupted or incomplete; however, they were accessible through alternative sources like university archives, ResearchGate, or authors' personal websites. Many articles lacked DOIs, and inconsistent formatting in Design Society conferences complicated reference management software, causing citation issues with Mendeley, RefWorks, and EndNote. These challenges highlight the need for the Design Society to standardize bibliographical formats across all conferences and publications for better accessibility and reference management.

Acknowledgements

This paper reflects our shared commitment to advancing how design and sustainability education engages with entrepreneurship. We thank each other for a deeply collaborative partnership - from co-developing the core idea, challenging assumptions through critical discussion, and iterating through multiple drafts - to bring this study to completion. With this paper we propose a starting point for further work and discussion on connecting design and entrepreneurship within the Design Society community.

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